# ResiDual for Audio: Spectral Reweighting of Residual Streams in CLAP Models

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**Lorenzo Arcioni**

## Abstract

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## 1. Method

## A. CLAP and HTS-AT: Architecture, Notation, and Residual Decomposition

This appendix is a self-contained reference for the two components at the core of our system. We begin by describing the CLAP dual-encoder model and its configuration (§A.1). We then present the HTS-AT audio backbone in detail, covering input preprocessing, hierarchical stage structure, and the internal mechanics of each attention block (§A.2). We then derive the per-head residual decomposition that underlies our analysis (§A.3). Finally, we provide implementation details for the ResiDual spectral reweighting (§C.1). All notation is introduced inline and collected in Table 3 for reference.

### A.1. CLAP Overview

CLAP (Contrastive Language–Audio Pretraining) (**?**) is a dual-encoder model that aligns audio and text representations in a shared embedding space via contrastive learning. The audio encoder is HTS-AT (**?**) and the text encoder is GPT-2 (**?**) (embedding dimension 768), whose weights are frozen throughout training. Both encoders are independently projected to a joint space of dimension $d_{\text{proj}} = 1024$ via dedicated linear projection heads; similarity is measured by temperature-scaled cosine similarity with InfoNCE loss at temperature $\tau = 0.003$. Zero-shot audio classification is performed by comparing an audio embedding against the embeddings of textual class prompts.

The full configuration used in this work is reported in Table 1.

*Table 1.* CLAP configuration parameters.

| Parameter | Value |
|---|---|
| Text encoder | GPT-2 |
| Text encoder embedding dim | 768 |
| Audio encoder | HTS-AT |
| Audio encoder output dim ($D_3$) | 768 |
| Joint projection dim ($d_{\text{proj}}$) | 1024 |
| Contrastive temperature $\tau$ | 0.003 |
| Sampling rate | 44 100 Hz |
| Audio duration | 7 s |
| Mel bands | 64 |
| FFT window | 1024 |
| Hop size | 320 |
| $f_{\min}$ / $f_{\max}$ | 50 Hz / 8 000 Hz |

### A.2. HTS-AT Architecture

HTS-AT (**?**) is a Swin-Transformer variant adapted for audio spectrograms. Its computation is organised into four hierarchical *stages*, each composed of several *blocks*, as illustrated in Figure 1. We describe the pipeline from raw audio to the final embedding in order: input preprocessing, patch embedding, stage structure, and the attention mechanism inside each block.

**Input preprocessing and patch embedding.** Each audio clip is converted to a 64-band log-mel spectrogram ($f_{\min} = 50$ Hz, $f_{\max} = 8{,}000$ Hz, STFT window 1024, hop 320) and normalised per mel-band. The resulting time–frequency matrix is rearranged into a square image $\mathbf{x}_{\text{img}} \in \mathbb{R}^{1\times 256\times 256}$ by folding the time axis into four contiguous segments of 256 frames stacked vertically over the 64 mel-frequency bands ($4\times 64 = 256$ rows, 256 columns). This image is then split into $4 \times 4$ non-overlapping patches via a strided `Conv2d` layer (kernel $4\times 4$, stride 4), followed

Email: Lorenzo Arcioni <arcioni.1885377@studenti.uniroma1.it>.

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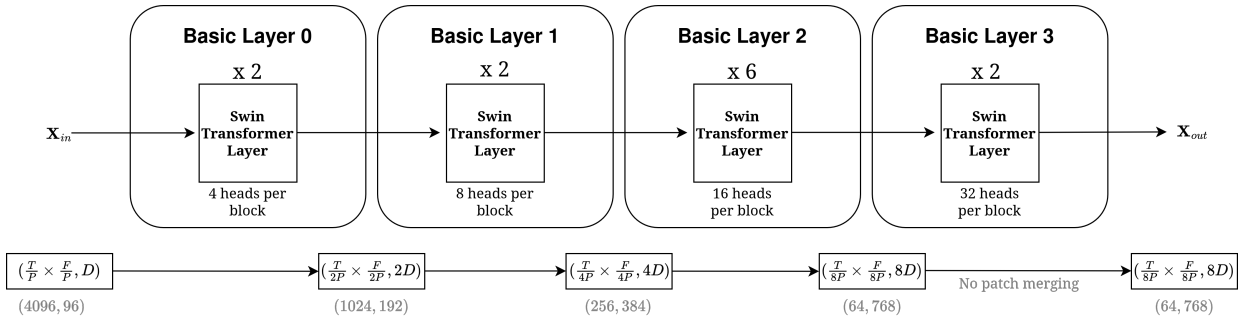


*Figure 1.* HTS-AT hierarchical architecture. The model consists of four stages of increasing embedding dimension: Stage 0 (2 blocks, 4 heads), Stage 1 (2 blocks, 8 heads), Stage 2 (6 blocks, 16 heads), and Stage 3 (2 blocks, 32 heads). PatchMerging is applied after Stages 0, 1, and 2, halving the spatial side length and doubling the feature dimension at each transition. Stage 3 has no PatchMerging. The input token sequence has spatial size $64 \times 64 = 4{,}096$ tokens with embedding dimension $D_0 = 96$; after three PatchMerging operations the final spatial size is $8 \times 8 = 64$ tokens with $D_3 = 768$.

by `LayerNorm`, yielding the input token sequence

$$\mathbf{Z}^{(0,0)} = \mathbf{X}_{\text{in}} \in \mathbb{R}^{4096\times 96}, \tag{1}$$

where $4096 = 64 \times 64$ tokens each have embedding dimension $D_0 = 96$.

**Stage structure and PatchMerging.** The four stages operate at progressively coarser spatial resolutions, as summarised in Table 2. We index the residual stream as $\mathbf{Z}^{(\ell,b)}$, where $\ell \in \{0, 1, 2, 3\}$ is the stage index and $b \in \{1, \ldots, B_\ell\}$ is the block index within that stage.

After Stages 0, 1, and 2, a *PatchMerging* layer concatenates each $2\times 2$ neighbourhood of spatially adjacent tokens into a single vector of dimension $4D_\ell$, then projects it to $2D_\ell$ via a bias-free linear layer. This halves the spatial side length $S_\ell$ and doubles the embedding dimension:

$$\mathbb{R}^{S_\ell^2 \times D_\ell} \xrightarrow{\text{PatchMerging}} \mathbb{R}^{(S_\ell/2)^2 \times 2D_\ell}. \tag{2}$$

Stage 3 has no PatchMerging. The resulting spatial side lengths are $S_\ell \in \{64, 32, 16, 8\}$ and the corresponding embedding dimensions are $D_\ell \in \{96, 192, 384, 768\}$ (see Table 2).

**Within-stage residual structure.** Inside each stage $\ell$, both the attention and MLP sub-layers write additively to the residual stream. Following the pre-norm formulation (**?**), the stream after block $b$ is:

$$\mathbf{Z}^{(\ell,b)} = \mathbf{Z}^{(\ell,0)} + \sum_{b'=1}^{b} \mathbf{A}_{\ell,b'} + \sum_{b'=1}^{b} \mathbf{M}_{\ell,b'}, \quad \in \mathbb{R}^{S_\ell^2 \times D_\ell}, \tag{3}$$

where $\mathbf{Z}^{(\ell,0)}$ is the stage input, $\mathbf{A}_{\ell,b}$ is the W-MSA output, $\mathbf{M}_{\ell,b}$ is the MLP output, and $S_\ell$ is the spatial side length of the token grid at stage $\ell$.

This decomposition holds strictly *within* a stage, where $D_\ell$ is constant. Across stage boundaries, PatchMerging changes both spatial resolution and embedding dimension ($S_\ell \to S_\ell/2$, $D_\ell \to 2D_\ell$), breaking any global additive structure across the full network, unlike in isotropic vision transformers (**?**).

**Per-head output of the attention layer.** The W-MSA output $\mathbf{A}_{\ell,b}$ is produced by concatenating the outputs of $H_\ell$ independent attention heads and projecting through a shared output matrix $W^O_{\ell,b}$. At block $(\ell, b)$, the raw output of head $h$ is (see Appendix A.2 for derivation):

$$\mathbf{H}_{\ell,b,h} = \text{Softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}} + \mathbf{B}_{\ell,b,h}\right)\mathbf{V}_h \;\in\; \mathbb{R}^{N_w^\ell \times M \times d_h}, \tag{4}$$

where $N_w^\ell = S_\ell^2/M$ is the number of attention windows, $M = 64$ tokens per window, and $\mathbf{B}_{\ell,b,h} \in \mathbb{R}^{M\times M}$ is the learned relative position bias for head $h$ at block $(\ell, b)$ (shared across windows, independent per head and per block). All $H_\ell$ head outputs are concatenated and projected to give the W-MSA output (see Appendix A.3 for the full decomposition):

$$\mathbf{A}_{\ell,b} = \left[\mathbf{H}_{\ell,b,1} \| \cdots \| \mathbf{H}_{\ell,b,H_\ell}\right] W^O_{\ell,b} + \mathbf{b}^O_{\ell,b} \;\in\; \mathbb{R}^{N_w^\ell \cdot M \times D_\ell}. \tag{5}$$

We analyse the raw head outputs $\mathbf{H}_{\ell,b,h}$ directly, *before* the output projection $W^O_{\ell,b}$. Each $\mathbf{H}_{\ell,b,h}$ lives in the per-head space $\mathbb{R}^{d_h}$, which is constant across all stages and blocks. This choice avoids entangling information across heads introduced by the shared projection, and enables uniform cross-stage comparisons of head geometry throughout the network. A complementary analysis of the projected contributions $\widehat{\mathbf{H}}_{\ell,b,h}$—which characterise how each head shapes the residual stream—is provided in Appendix **??**.

**Block computation: W-MSA and MLP.** Each block $b$ at stage $\ell$ applies two sub-layers in sequence, both preceded

*Table 2.* HTS-AT stage-level architectural parameters. The per-head dimension $d_h = D_\ell/H_\ell = 24$ is constant across all stages. $S_\ell$ is the spatial side length of the token grid at stage $\ell$; $N_w^\ell = S_\ell^2/M$ is the number of attention windows, with $M = w^2 = 64$ tokens per window ($w = 8$).

| **Stage** $\ell$ | **Blocks** $B_\ell$ | **Heads** $H_\ell$ | **Dim** $D_\ell$ | **Spatial** $S_\ell^2$ | **Windows** $N_w^\ell$ |
|---|---|---|---|---|---|
| 0 | 2 | 4 | 96 | $64 \times 64$ | 64 |
| 1 | 2 | 8 | 192 | $32 \times 32$ | 16 |
| 2 | 6 | 16 | 384 | $16 \times 16$ | 4 |
| 3 | 2 | 32 | 768 | $8 \times 8$ | 1 |
| Total heads $H_{\text{tot}}$ | $2{\cdot}4 + 2{\cdot}8 + 6{\cdot}16 + 2{\cdot}32 = 8 + 16 + 96 + 64 = 184$ | | | | |

by LayerNorm and connected via residual additions:

$$\mathbf{Z}^{(\ell,b)} \leftarrow \mathbf{Z}^{(\ell,b-1)} + \text{W-MSA}_{\ell,b}\Big(\text{LN}\Big(\mathbf{Z}^{(\ell,b-1)}\Big)\Big), \quad (6)$$

$$\mathbf{Z}^{(\ell,b)} \leftarrow \mathbf{Z}^{(\ell,b)} + \text{MLP}_{\ell,b}\Big(\text{LN}\Big(\mathbf{Z}^{(\ell,b)}\Big)\Big). \quad (7)$$

Even-indexed blocks use standard Window Multi-head Self-Attention (W-MSA); odd-indexed blocks use Shifted-Window MSA (SW-MSA), which shifts the partition by $(w/2, w/2)$ tokens to enable cross-window interactions. At Stage 3, where $S_3 = w = 8$ so the grid coincides with a single window, the shift degenerates to zero and all blocks use W-MSA. The MLP sub-layer is a two-layer feed-forward network with hidden dimension $4D_\ell$ and GELU activation.

**WindowAttention in detail.** The W-MSA module at block $(\ell, b)$ first partitions the $S_\ell^2$ tokens into $N_w^\ell = S_\ell^2/M$ non-overlapping windows of $M = 64$ tokens each, then applies multi-head attention independently within each window.

In the SW-MSA variant, a cyclic shift is applied before window partitioning.

Queries, keys, and values are computed via a single fused projection $W_{\ell,b}^{QKV} \in \mathbb{R}^{D_\ell \times 3D_\ell}$:

$$\text{LN}(\mathbf{Z}^{(\ell,b-1)})\, W_{\ell,b}^{QKV} \xrightarrow{\text{split}(D_\ell)} \mathbf{Q},\, \mathbf{K},\, \mathbf{V} \;\in\; \mathbb{R}^{N_w^\ell \times M \times D_\ell}. \quad (8)$$

Each tensor is then reshaped to isolate the $H_\ell$ heads, yielding per-head slices $\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h \in \mathbb{R}^{N_w^\ell \times M \times d_h}$ with $d_h = D_\ell/H_\ell = 24$.

For each head $h \in \{1, \ldots, H_\ell\}$, attention is computed as:

$$\mathbf{H}_{\ell,b,h} = \text{Softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}} + \mathbf{B}_{\ell,b,h}\right)\mathbf{V}_h \;\in\; \mathbb{R}^{N_w^\ell \times M \times d_h}, \quad (9)$$

where $\mathbf{B}_{\ell,b,h} \in \mathbb{R}^{M \times M}$ is the learned relative position bias for head $h$ of block $(\ell, b)$. This bias encodes the relative spatial offset between each pair of tokens *within* a window; it is shared across all $N_w^\ell$ windows (broadcast) and is independent for each head. Concretely, it is read from a parameter table of shape $((2w-1)^2, H_\ell) = (225, H_\ell)$ via a fixed index mapping, so each head $h$ has its own column of learnable values.

All $H_\ell$ head outputs are then concatenated and passed through the block-specific output projection $W_{\ell,b}^O \in \mathbb{R}^{D_\ell \times D_\ell}$ with bias $\mathbf{b}_{\ell,b}^O \in \mathbb{R}^{D_\ell}$, giving the W-MSA output

$$\mathbf{A}_{\ell,b} = \big[\mathbf{H}_{\ell,b,1} \| \cdots \| \mathbf{H}_{\ell,b,H_\ell}\big]\, W_{\ell,b}^O + \mathbf{b}_{\ell,b}^O \;\in\; \mathbb{R}^{N_w^\ell \cdot M \times D_\ell}, \quad (10)$$

which is added to the residual stream in Eq. (17).

### A.3. Per-Head Residual Decomposition

The additive residual structure of Eq. (17) allows us to decompose the W-MSA contribution $\mathbf{A}_{\ell,b}$ exactly into independent per-head terms. Denoting by $W_{\ell,b,h}^O \in \mathbb{R}^{d_h \times D_\ell}$ the row slice of $W_{\ell,b}^O$ corresponding to head $h$ (rows $[(h-1)d_h,\ hd_h)$), we distribute the output projection over heads:

$$\begin{aligned}\mathbf{A}_{\ell,b} &= \big[\mathbf{H}_{\ell,b,1} \| \cdots \| \mathbf{H}_{\ell,b,H_\ell}\big] W_{\ell,b}^O + \mathbf{b}_{\ell,b}^O \\ &= \sum_{h=1}^{H_\ell}\Big(\mathbf{H}_{\ell,b,h}\, W_{\ell,b,h}^O + \tfrac{\mathbf{b}_{\ell,b}^O}{H_\ell}\Big) \;=\; \sum_{h=1}^{H_\ell} \widehat{\mathbf{H}}_{\ell,b,h}, \quad (11)\end{aligned}$$

where we define the *per-head projected contribution*

$$\widehat{\mathbf{H}}_{\ell,b,h} \;=\; \mathbf{H}_{\ell,b,h}\, W_{\ell,b,h}^O + \frac{\mathbf{b}_{\ell,b}^O}{H_\ell} \;\in\; \mathbb{R}^{N_w^\ell \times M \times D_\ell}. \quad (12)$$

This decomposition is exact and follows from the linearity of matrix multiplication: because $W_{\ell,b}^O$ acts on the concatenation of all head outputs, each head $h$ effectively multiplies only its own row slice $W_{\ell,b,h}^O$, and the output bias can be distributed equally among heads without any approximation.

The decomposition holds within a single stage, where $D_\ell$ is constant. Across stage boundaries, PatchMerging changes both spatial resolution and embedding dimension, breaking any global additive structure; cross-stage comparisons must therefore account for this.

# B. Head Analisys

## B.1. Spatial Aggregation and Dataset Representations

Each raw head output $\mathbf{H}_{\ell,b,h}$ is a three-dimensional tensor indexed by window, token position, and feature dimension. To obtain a single fixed-size vector per audio sample, we mean-pool over all windows and token positions:

$$\mathbf{r}_{\ell,b,h} = \frac{1}{N_w^\ell \, M} \sum_{i=1}^{N_w^\ell} \sum_{j=1}^{M} \mathbf{H}_{\ell,b,h}[i,j,:] \;\in\; \mathbb{R}^{d_h}. \tag{13}$$

This yields one vector $\mathbf{r}_{\ell,b,h}$ per audio sample, per head $h$, at every block $(\ell, b)$ of the network. Stacking these vectors across the $n$ samples in the dataset gives the matrix

$$\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}, \tag{14}$$

which is the primary object of our analysis. The total number of such matrices across the model is $H_{\text{tot}} = 184$ (one per head per block; see Table 2).

Since $d_h = 24$ is constant across all stages, the ambient dimension of $\mathbf{R}_{\ell,b,h}$ is uniform throughout the network, enabling direct cross-stage comparisons of all dimensionality metrics without correction for a varying ceiling.

**Dataset sampling.** We extract representations from four audio classification benchmarks:

- **ESC-50** (**?**): 50 environmental sound classes, 2,000 clips (5 s, 44.1 kHz).
- **TinySOL** (**?**): 14 orchestral instrument classes with varied articulations, 2,071 monophonic samples (1–16 s, 44.1 kHz).
- **VocalSound** (**?**): 6 non-speech vocal categories, stratified subset of 1,200 samples.
- **IRMAS** (**?**): 11 musical instrument classes, annotated audio excerpts with predominant instrument.

Audio preprocessing follows the CLAP standard pipeline: 64-band log-mel spectrogram ($f_{\min} = 50\,\text{Hz}$, $f_{\max} = 8000\,\text{Hz}$, FFT window 1024, hop 320), padded or truncated to 7 seconds at 44.1 kHz.

## B.2. Intrinsic Dimensionality Analysis

To characterize the effective complexity of the projected head representations $\{\mathbf{r}_{\ell,b,h}^{(i)}\}_{i=1}^{n} \subset \mathbb{R}^{1024}$, we employ a battery of linear and nonlinear dimensionality estimators.

### B.2.1. Linear Estimators

**PCA-based dimensionality.** For each head, let $\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}$ stack all aggregated representations. We compute the covariance matrix $\mathbf{C}_{\ell,b,h} = \frac{1}{n-1}\mathbf{R}^\top \mathbf{R}$ and obtain ordered eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots$. Linear intrinsic dimensionality is:

$$d_{\text{PCA}}(\alpha) = \arg\min_k \left\{ \frac{\sum_{i=1}^{k} \lambda_i}{\sum_i \lambda_i} \geq \alpha \right\}, \tag{15}$$

evaluated at $\alpha \in \{0.90, 0.95, 0.99\}$. We also report the *Explained Variance Ratio* of the first principal component, $\text{EVR}_1 = \lambda_1 / \sum_i \lambda_i$.

**Participation Ratio.**

$$\text{PR} = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}. \tag{16}$$

High PR signals uniform variance distribution; low PR signals dominance of few directions.

**Effective Rank.**

$$\text{EffRank} = \exp\left( -\sum_i p_i \log p_i \right), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j}, \tag{17}$$

which can be interpreted as the exponential of the spectral entropy. Unlike PCA thresholds, it provides a smooth, threshold-free notion of dimensionality, capturing how uniformly variance is distributed across components.

### B.2.2. Nonlinear Estimators

**TwoNN.**

$$d_{\text{TwoNN}} = \left( \frac{1}{n} \sum_{i=1}^{n} \log \frac{r_2^{(i)}}{r_1^{(i)}} \right)^{-1}, \tag{18}$$

where $r_1^{(i)}, r_2^{(i)}$ are Euclidean distances to the first and second nearest neighbours of $\mathbf{r}_{\ell,b,h}^{(i)}$ (**?**).

**MLE.**

$$\hat{d}_{\text{MLE}}(\mathbf{r}) = \left( \frac{1}{k-1} \sum_{j=1}^{k-1} \log \frac{r_k(\mathbf{r})}{r_j(\mathbf{r})} \right)^{-1}, \tag{19}$$

averaged over all samples (**?**), with $k = 20$. This estimator is based on a local maximum likelihood principle, assuming uniform sampling in small neighbourhoods, and is sensitive to fine-scale manifold structure beyond global linear approximations.

**Linear-Nonlinear (L/N) Ratio**

$$\text{Ratio}_B = \frac{\bar{d}_{\text{PCA}_{99}}}{\bar{d}_{\text{TwoNN}}}. \tag{20}$$

Values near 1 indicate near-linear manifolds; higher values signal nonlinear curvature beyond what PCA captures, and serve as a diagnostic for selecting layer targets for spectral reweighting.

# C. Reweighting Pipeline

## C.1. ResiDual: Implementation Details

This section describes the full implementation of ResiDual*(RD*), the learnable variant of ResiDual in which the spectral reweighting weights are optimised end-to-end on a held-out validation set rather than set by a closed-form formula. The pipeline has two sequential phases: a *fitting pass* that estimates a fixed PCA basis for each attention head (§C.1.1), followed by an *optimisation phase* in which the per-head weight vectors are treated as free parameters and updated via gradient descent to maximise zero-shot classification accuracy (§C.1.3). The forward-pass reweighting that uses these weights at inference time is described in §C.1.2.

**Module injection via forward hooks.** RD* injects its reweighting through PyTorch forward hooks. For each target block $(\ell, b)$ with $\ell \in \mathcal{L}$, a dedicated `AttentionHook` object is registered on the corresponding `WindowAttention` module immediately before the forward pass begins and removed as soon as the forward pass for that layer completes.

The hook intercepts the tuple $(\mathbf{x}_{\text{out}}, \mathbf{A})$ returned by `WindowAttention`, where $\mathbf{x}_{\text{out}} \in \mathbb{R}^{B' \times N \times C}$ is the projected attention output and $\mathbf{A}$ contains the attention weights. It then reconstructs the per-head pre-projection tensors $\mathbf{H}_{\ell,b,h}$ by re-applying the frozen QKV projection to the hook input, isolating the value slice $\mathbf{V}_h$, and computing $\mathbf{x}_{\text{heads}} = \mathbf{A}\,\mathbf{V}$. This yields tensors of shape $(B', H_\ell, N, d_h)$ without any additional trainable operation. No structural modification of the original model is performed: all pretrained parameters — the fused QKV projection $W^{QKV}_{\ell,b}$, the output projection $W^{O}_{\ell,b}$ with bias $\mathbf{b}^{O}_{\ell,b}$, and the relative position bias table — remain exactly as loaded from the checkpoint. The only additions are the learnable weight vectors $\boldsymbol{\lambda}_{\ell,b,h}$ and their associated PCA buffers, stored in a separate `nn.ModuleDict` that is indexed by $(\ell, b, h)$.

### C.1.1. Fitting Pass: PCA Basis Estimation

**Data collection.** The fitting pass is run with gradients disabled and the model in evaluation mode. For each audio clip (up to $N_{\max}$ clips), a forward pass is executed with data collection enabled. The hook registered at each target block intercepts, for every head $h$, the pre-reweighting tensor $\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{B' \times H_\ell \times N \times d_h}$ and flattens it along the spatial and batch axes to obtain $\mathbb{R}^{(B' \cdot N) \times d_h}$ before appending it to an accumulation buffer for $(\ell, b, h)$. Here $B'$ is the number of attention windows times the batch size and $N = M = 64$ is the number of tokens per window. After all clips have been processed, the buffers are concatenated to form the fitting matrix

$$\widetilde{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{N_{\text{fit}} \times d_h}, \qquad N_{\text{fit}} = \sum_{\text{batches}} B \cdot N^{\ell}_{w} \cdot M, \quad (21)$$

where the summation is over all processed batches of size $B$.

**PCA decomposition.** Standard PCA is applied to $\widetilde{\mathbf{R}}_{\ell,b,h}$: the empirical mean $\boldsymbol{\mu}_{\ell,b,h} \in \mathbb{R}^{d_h}$ is computed and subtracted, and the empirical covariance is decomposed as

$$\begin{aligned}\mathbf{C}_{\ell,b,h} &= \frac{1}{N_{\text{fit}} - 1}\left(\widetilde{\mathbf{R}}_{\ell,b,h} - \mathbf{1}\boldsymbol{\mu}^{\top}_{\ell,b,h}\right)^{\top}\left(\widetilde{\mathbf{R}}_{\ell,b,h} - \mathbf{1}\boldsymbol{\mu}^{\top}_{\ell,b,h}\right) \\ &= \boldsymbol{\Phi}_{\ell,b,h}\,\operatorname{diag}(\lambda^{\text{ev}}_1, \ldots, \lambda^{\text{ev}}_{d_h})\,\boldsymbol{\Phi}^{\top}_{\ell,b,h}, \quad \lambda^{\text{ev}}_1 \geq \cdots \geq \lambda^{\text{ev}}_{d_h} \geq 0,\end{aligned} \quad (22)$$

where $\boldsymbol{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$ is the orthogonal matrix of eigenvectors and $\lambda^{\text{ev}}_j$ denotes the $j$-th eigenvalue of the empirical covariance (we use the superscript "ev" to distinguish these fixed eigenvalues from the trainable weight vector $\boldsymbol{\lambda}_{\ell,b,h}$ introduced below). The full basis $\boldsymbol{\Phi}_{\ell,b,h}$ and the mean $\boldsymbol{\mu}_{\ell,b,h}$ are stored as non-trainable buffers; the temporary accumulation buffers are then cleared to free memory.

**Task-relevant subspace and initialisation of $\boldsymbol{\lambda}$.** For each head, the index

$$k_{\ell,b,h} = \min\left\{ j \in \{1, \ldots, d_h\} : \sum_{i=1}^{j} \frac{\lambda^{\text{ev}}_i}{\sum_{i'} \lambda^{\text{ev}}_{i'}} \geq \rho_{\text{var}} \right\} \quad (23)$$

identifies the smallest number of principal components that explain at least a fraction $\rho_{\text{var}}$ of the total variance, providing a per-head estimate of the effective dimensionality of the task-relevant subspace. The learnable weight vector $\boldsymbol{\lambda}_{\ell,b,h} \in \mathbb{R}^{d_h}$ is then initialised as

$$\lambda^{(0)}_j = \begin{cases} 1 & j \leq k_{\ell,b,h}, \\ 0 & j > k_{\ell,b,h}. \end{cases} \quad (24)$$

This initialisation corresponds to the identity map restricted to the task-relevant subspace and a complete suppression of the noise components: before any optimisation, RD*acts as a projection-and-reconstruction operator that retains the top-$k$ PCA directions unchanged and zeros out the remainder.

### C.1.2. Forward Pass with Spectral Reweighting

At inference time, the hook at block $(\ell, b)$ intercepts the per-head tensors $\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{B' \times N \times d_h}$ and applies the following sequence of operations for each head $h$.

**Step 1 — Centering.**

$$\widetilde{\mathbf{H}}_{\ell,b,h} \;=\; \mathbf{H}_{\ell,b,h} - \boldsymbol{\mu}_{\ell,b,h} \;\in\; \mathbb{R}^{B' \times N \times d_h}, \tag{25}$$

where $\boldsymbol{\mu}_{\ell,b,h}$ is broadcast over the first two axes.

**Step 2 — Projection onto the full PCA basis.**

$$\mathbf{Z}_{\ell,b,h} \;=\; \widetilde{\mathbf{H}}_{\ell,b,h}\, \boldsymbol{\Phi}_{\ell,b,h} \;\in\; \mathbb{R}^{B' \times N \times d_h}. \tag{26}$$

Unlike fixed-weight variants that truncate to $K < d_h$ components, RD*projects onto the *full* basis $\boldsymbol{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$, delegating the effective truncation to the learned weights.

**Step 3 — Per-component rescaling by $\boldsymbol{\lambda}_{\ell,b,h}$.**

$$\widehat{\mathbf{Z}}_{\ell,b,h} \;=\; \mathbf{Z}_{\ell,b,h} \odot \boldsymbol{\lambda}_{\ell,b,h}^{\top} \;\in\; \mathbb{R}^{B' \times N \times d_h}, \tag{27}$$

where $\boldsymbol{\lambda}_{\ell,b,h}^{\top}$ is broadcast over the first two axes. Because $\boldsymbol{\lambda}_{\ell,b,h}$ is a free real-valued parameter, each entry $\lambda_j$ can amplify ($|\lambda_j| > 1$), attenuate ($|\lambda_j| \in (0,1)$), suppress ($\lambda_j = 0$), or invert ($\lambda_j < 0$) the corresponding principal component.

**Step 4 — Reconstruction in the original space.**

$$\mathbf{H}^{*}_{\ell,b,h} \;=\; \widehat{\mathbf{Z}}_{\ell,b,h}\, \boldsymbol{\Phi}_{\ell,b,h}^{\top} \;\in\; \mathbb{R}^{B' \times N \times d_h}. \tag{28}$$

The empirical mean is *not* restored after reconstruction. Since $\mathbf{H}^{*}_{\ell,b,h}$ is immediately passed to the output projection $W^{O}_{\ell,b}$, which has its own learnable bias $\mathbf{b}^{O}_{\ell,b}$, re-adding $\boldsymbol{\mu}_{\ell,b,h}$ would introduce a systematic, data-dependent shift in the residual stream that is already absorbed by $\mathbf{b}^{O}_{\ell,b}$.

Combining Steps 1–4, the full RD*transform can be written compactly as

$$\mathrm{RD}^{*}(\mathbf{H}_{\ell,b,h},\, \boldsymbol{\lambda}_{\ell,b,h}) \;=\; \boldsymbol{\Phi}_{\ell,b,h}\, \mathrm{diag}(\boldsymbol{\lambda}_{\ell,b,h})\, \boldsymbol{\Phi}_{\ell,b,h}^{\top} \big(\mathbf{H}_{\ell,b,h} - \boldsymbol{\mu}_{\ell,b,h}\big). \tag{29}$$

The tensor $\mathbf{H}^{*}_{\ell,b,h}$ replaces $\mathbf{H}_{\ell,b,h}$ in Eq. (21), so the output projection $W^{O}_{\ell,b}$ and the residual-stream update of Eq. (17) proceed unchanged. The gradient of any downstream loss flows through $\mathrm{RD}^{*}$ and updates only $\boldsymbol{\lambda}_{\ell,b,h}$; the basis $\boldsymbol{\Phi}_{\ell,b,h}$, the mean $\boldsymbol{\mu}_{\ell,b,h}$, and all original model parameters are held fixed.

### C.1.3. Optimisation of $\boldsymbol{\lambda}$ via Zero-Shot Accuracy

After the fitting pass has fixed the PCA bases and initialised $\boldsymbol{\lambda}$ via Eq. (27), the second phase optimises the weight vectors to maximise zero-shot classification accuracy on a held-out validation set.

**Parameter isolation.** All model parameters except the weight vectors $\{\boldsymbol{\lambda}_{\ell,b,h}\}_{\ell \in \mathcal{L},\, b,\, h}$ are frozen by setting `requires_grad = False`. This ensures that the only degrees of freedom during optimisation are the $|\mathcal{L}| \times \sum_{\ell \in \mathcal{L}} B_\ell \cdot H_\ell$ scalar values that control the per-component gain of each attention head in the target stages. The original CLAP audio projection head $P : \mathbb{R}^{D_3} \to \mathbb{R}^{d_{\text{proj}}}$ is kept frozen but is included in the computational graph so that gradients flow from the similarity space back to $\boldsymbol{\lambda}$.

**Objective.** Given a batch of audio clips with ground-truth class labels $y \in \{1, \ldots, C\}$ and pre-computed, $\ell_2$-normalised text embeddings $\mathbf{t}_c \in \mathbb{R}^{d_{\text{proj}}}$ for each class $c$, the optimisation objective is the cross-entropy loss over zero-shot logits:

$$\mathcal{J}(\boldsymbol{\lambda}) \;=\; -\,\mathbb{E}_{(x,y)} \left[ \log \frac{\exp\big(\hat{\mathbf{a}}(x)^{\top} \mathbf{t}_y\big)}{\sum_{c=1}^{C} \exp\big(\hat{\mathbf{a}}(x)^{\top} \mathbf{t}_c\big)} \right], \tag{30}$$

where $\hat{\mathbf{a}}(x) = \ell_2\text{-norm}\big(P(\mathrm{latent}(x))\big) \in \mathbb{R}^{d_{\text{proj}}}$ is the $\ell_2$-normalised audio embedding of clip $x$ and $\mathrm{latent}(x)$ is the spatially averaged final-stage output of HTS-AT after applying the RD*transforms. Minimising $\mathcal{J}$ is equivalent to maximising zero-shot accuracy in expectation.

**Optimiser and schedule.** The weight vectors are updated with Schedule-Free Adam (**?**) at learning rate $\eta$, which eliminates the need for a separate learning-rate schedule while matching or exceeding the performance of Adam with cosine decay on this type of fine-tuning task. When the `schedulefree` package is unavailable, standard Adam is used as a fallback.

**Early stopping and best-$\boldsymbol{\lambda}$ restoration.** At the end of each epoch, zero-shot accuracy on the validation set is computed. A snapshot of the current $\boldsymbol{\lambda}$ vectors is saved whenever a new accuracy maximum is achieved. Training halts early if no improvement is observed for $p$ consecutive epochs. At termination, the best snapshot is restored, guaranteeing that the final weight vectors correspond to the configuration that achieved the highest observed validation accuracy rather than to the last iterate.

**Hyperparameter summary.** Table 4 summarises all hyperparameters introduced by RD*and the default values used in all experiments reported in this paper.

**Hook-based extraction (updated note).** The hook-based mechanism described in §1.2 for the dimensionality analysis is also the mechanism through which RD*applies reweighting during inference. The same `AttentionHook` object operates in two distinct modes depending on the `collect_for_fitting` flag: in collection mode it accumulates raw head activations for PCA estimation without modifying the forward pass output; in reweighting mode it replaces the projected attention output with the RD*-transformed version before returning it to the residual stream. Crucially, in both modes the hooks

are *temporary*: they are installed at the start of each call to `_forward_layer_with_attention_hooks` and removed immediately upon its completion, so the underlying `WindowAttention` modules are never structurally modified.

## D. Appendice Ultima

**Approximation scope and token-level reweighting.** The mean-pooled representation $\mathbf{r}_{\ell,b,h}$ is used exclusively for the dimensionality analysis reported in this section. Pooling over all $N_w^\ell \cdot M$ spatial positions collapses the intra-sample token variance, retaining only the inter-sample variance that drives the PCA basis estimation. This choice is motivated by computational constraints: constructing the full token-level matrix $\widetilde{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{n\, N_w^\ell M \times d_h}$ for all 184 heads simultaneously would exceed available memory (considering the all linear and non linear estimators). Despite this approximation, the pooled statistics provide a faithful characterisation of the dominant geometric structure of each head's subspace, since the inter-sample variance captures the directions along which heads discriminate between audio inputs—which is precisely the quantity relevant to understanding head specialisation.

The spectral reweighting in ResiDual, as specified in the next sections, operates instead at full spatial resolution.This token-level granularity allows ResiDual to modulate the geometry of individual spatial positions rather than acting on a global summary, making the intervention more targeted without sacrificing the precision of the basis estimation.

**Hook-based extraction.** To obtain $\mathbf{H}_{\ell,b,h}$ without modifying the model, we register forward hooks on the `WindowAttention` module of each block $(\ell, b)$. The hook intercepts the tensor `(attn @ v)` immediately after the per-head attention computation of Eq. (15), yielding the raw outputs $\mathbf{H}_{\ell,b,h}$ of shape $(N_w^\ell,\ M,\ d_h)$. No further projection is applied: the hook output is spatially pooled directly via Eq. (4) to produce $\mathbf{r}_{\ell,b,h}$.

*Table 3.* Summary of notation used throughout the paper. Symbols introduced by RD$^*$ are marked with $(\star)$.

| **Symbol** | **Definition** |
|---|---|
| $\ell \in \{0, 1, 2, 3\}$ | Stage index |
| $b \in \{1, \ldots, B_\ell\}$ | Block index within stage $\ell$ |
| $h \in \{1, \ldots, H_\ell\}$ | Attention head index within block $(\ell, b)$ |
| $S_\ell$ | Spatial side length of the token grid at stage $\ell$ |
| $w = 8$ | Attention window side length |
| $M = w^2 = 64$ | Tokens per attention window |
| $N_w^\ell = S_\ell^2 / M$ | Number of attention windows at stage $\ell$ |
| $d_h = 24$ | Per-head feature dimension (constant across stages) |
| $D_\ell = H_\ell \cdot d_h$ | Total embedding dimension at stage $\ell$ |
| $d_{\text{proj}} = 1024$ | CLAP joint embedding dimension |
| $W_{\ell,b}^{QKV} \in \mathbb{R}^{D_\ell \times 3D_\ell}$ | Fused QKV projection at block $(\ell, b)$ |
| $W_{\ell,b}^{O} \in \mathbb{R}^{D_\ell \times D_\ell}$ | Output projection at block $(\ell, b)$ |
| $W_{\ell,b,h}^{O} \in \mathbb{R}^{d_h \times D_\ell}$ | Row slice of $W_{\ell,b}^{O}$ for head $h$ |
| $\mathbf{b}_{\ell,b}^{O} \in \mathbb{R}^{D_\ell}$ | Output projection bias at block $(\ell, b)$ |
| $\mathbf{B}_{\ell,b,h} \in \mathbb{R}^{M \times M}$ | Relative position bias for head $h$ at block $(\ell, b)$ |
| $\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$ | Raw head output at block $(\ell, b)$, head $h$ (pre-$W_{\ell,b}^{O}$) |
| $\widehat{\mathbf{H}}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times D_\ell}$ | Per-head projected contribution (post-$W_{\ell,b}^{O}$); see Eq. (23) |
| $\mathbf{r}_{\ell,b,h} \in \mathbb{R}^{d_h}$ | Spatially aggregated raw head output; see Eq. (4) |
| $\widehat{\mathbf{r}}_{\ell,b,h} \in \mathbb{R}^{D_\ell}$ | Spatially aggregated projected head output; see Eq. (**??**) |
| $\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}$ | Dataset matrix of $\mathbf{r}_{\ell,b,h}$ across $n$ samples |
| $\widehat{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{n \times D_\ell}$ | Dataset matrix of $\widehat{\mathbf{r}}_{\ell,b,h}$ across $n$ samples |
| $P : \mathbb{R}^{768} \to \mathbb{R}^{1024}$ | CLAP audio projection head (two-layer MLP with residual) |
| $n$ | Number of audio samples in the dataset |
| *Symbols introduced by RD$^*$ $(\star)$* | |
| $\mathcal{L} \subseteq \{0, 1, 2, 3\}$ $(\star)$ | Target stage indices for reweighting |
| $\rho_{\text{var}} \in (0, 1)$ $(\star)$ | Variance threshold for per-head subspace estimation (Eq. (26)) |
| $k_{\ell,b,h} \in \{1, \ldots, d_h\}$ $(\star)$ | Number of task-relevant PCA components for head $h$ at block $(\ell, b)$, determined by $\rho_{\text{var}}$ (Eq. (26)) |
| $\boldsymbol{\lambda}_{\ell,b,h} \in \mathbb{R}^{d_h}$ $(\star)$ | Learnable per-component weight vector for head $h$ at block $(\ell, b)$; initialised via Eq. (27), optimised via Eq. (3 |
| $\mathbf{C}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$ | Empirical covariance matrix of $\widetilde{\mathbf{R}}_{\ell,b,h}$ |
| $\boldsymbol{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$ | Orthogonal matrix of eigenvectors of $\mathbf{C}_{\ell,b,h}$ |
| $\lambda_j^{\text{ev}}$ | $j$-th eigenvalue of $\mathbf{C}_{\ell,b,h}$, $\lambda_1^{\text{ev}} \geq \cdots \geq \lambda_{d_h}^{\text{ev}} \geq 0$ |

*Table 4.* RD$^*$ hyperparameters.

| Symbol | Description | Default |
|---|---|---|
| $\mathcal{L}$ | Target stage indices for reweighting | $\{1, 2, 3\}$ |
| $\rho_{\text{var}}$ | Variance threshold for $k$ estimation (Eq. (26)) | $0.95$ |
| $\eta$ | Learning rate for Schedule-Free Adam | $10^{-2}$ |
| $p$ | Early-stopping patience (epochs) | $5$ |
| $T_{\max}$ | Maximum number of optimisation epochs | $30$ |
| $N_{\max}$ | Maximum fitting samples | $10\,000$ |